

CariSurg MedTech Pathways Research Project

Preliminary Proposal

Project Title: *Developing a Validated AI-Based Triage Decision-Support Tool for Emergency Departments: A Preliminary Pilot at Mercer General Hospital*

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Statement of Problem

Triage nurses at Mercer General Hospital face increasing pressure from seasonal surges, staffing shortages and rising emergency department (ED) demand. Current systems, such as CTAS and MTS rely on clinical judgement and can result in undertriage or overtriage in approximately one in every six assessments Porto. (1). In addition, Caribbean emergency departments lack validated AI decision support tools tailored to local populations. This 12-week pilot will assess whether a machine learning (ML) model trained on Mercer ED data can reduce triage inconsistencies.

Previous Work - Literature Review

The following review examines eight studies to establish what is known about AI-assisted triage, where existing models perform well and critical gaps remain. The literature collectively demonstrates that while ML models can outperform traditional triage scales in controlled settings, questions regarding the validity, algorithmic bias and clinical trust remain unresolved, particularly in resource-limited environments such as Caribbean EDs.

Sittiprawiat et al. (2) provides the clearest evidence that ML models can outperform traditional triage systems in predicting critical patient outcomes. The study used 163,452 ED visits from a Thai hospital (2018-2022) and tested XGBoost, Logistic Regression and Random Forest against CTAS. The study also processed patient complaints via a multilingual sentence embedding, a natural language processing (NLP) technique used to convert text into numerical representations. XGBoost outperformed CTAS with an AUROC of 0.917 versus 0.882, with the mode of arrival, age and patient complaints being the strongest predictors. However, the lack of patients who deteriorated post admission may have overestimated model performance and a

single institution training limits its applicability to other clinical settings including Caribbean EDs.

Porto. (1) whose Preferred Reporting Items for Systematic Reviews and Meta-Analyses (PRISMA)-complaint systematic review of 60 studies that covered 57 ML algorithms provides the largest evidence in this review and addresses whether this performance is consistent across various datasets and algorithms. XGBoost, Gradient Boosting and Deep Neural Networks achieved the highest performance with NLP techniques further improving triage classifications across patient complaints and clinical notes. The strongest predictors were oxygen saturation, patient complaints, systolic blood pressure, age and mode of arrival. Porto. (1) found that all validated models were developed using data from Asia, Europe or North America with no studies identified using Caribbean patient data, establishing the primary gap this pilot addresses.

While Rockenschaub et al. (3) focused on ICU scoring systems, the finding that external validation consistently reduces model performance has broad implications for any AI clinical tools used outside of its training environment, including ED triage. This review challenges the notion that strong performance in one setting will transfer to another. According to the review, external validation is rare in the ML literature. When external validation is done, models typically perform worse with an AUROC reduction seen in 49.5% of studies. This raises concerns about the dependability of models that have only been tested with their own training populations. This finding strengthens the methodological case for training and validating a model specifically on Mercer ED data, rather than attempting to apply an existing model.

Strong predictive performance alone does not guarantee safe clinical deployment. Yang et al. (4) moved beyond describing the bias as a problem and proposed a technical structure for actively mitigating it by introducing a learning approach that is designed to reduce ethnicity-based bias that the models acquire from data collection. Evaluated on COVID-19 screenings in the EDs, the framework was able to achieve effective screening performance while also slightly improving outcome fairness compared to standard ML benchmarks. This demonstrated that accuracy and fairness are not mutually exclusive. Yang et al. (4)'s primary limitation is that it was developed and validated in the UK and its learning reinforcement approach may require computational resources that exceed the scope of a 12-week pilot.

Adam et al. (5) shifts the focus from technical bias to behavioural risk of using AI outputs in high-risk clinical settings. Using experimental methods, Adam et al. (5) exposed

clinicians and non-experts to bias the algorithm outputs and compare their responses to prescriptive recommendations and descriptive flags. The biased AI recommendations influenced the decision-making for both groups, especially in cases that involved racial or religious minorities. The study's main limitation was that it used simulated emergency cases rather than real clinical environments that limits the direct application to live ED settings.

Hinson et al. (6) investigated whether greater clarity in AI outputs can either improve or complicate clinical decision-making. Hinson et al. (6) evaluated the outcomes of an AI triage tool that was implemented across multiple US EDs to assess its effect on triage equity. The study discovered that traditional ED triage are prone to high interoperator variability and has been linked to biased decision making based on race, ethnicity and spoken language. The AI-informed decision support demonstrated promise in reducing these disparities at implementation sites, however, the study lacked detail in methodology and outcomes, limiting the depth of the conclusions that can be drawn. Despite this, it provides real-world implementation context that is largely absent in other literatures and directly supports the argument that output design and equity must be considered alongside predictive accuracy.

Rosenbacke et al. (7) further expanded the question of how AI outputs are communicated to clinicians by investigating whether making AI more explainable improves clinical decision-making. Rosenbacke et al. (7) used a PRISMA-complaint systematic review of PubMed, Web of Science and other databases, which included studies that measured trust as an outcome of Explainable Artificial Intelligence (XAI) in clinical settings. While most studies found that XAI increased trust, the review discovered that overconfidence led some clinicians to defer to AI outputs even when they contradicted sound clinical judgement, implying that explainability and trust did not have a direct linear relationship. Comparison between studies were limited by inconsistent definitions of trust and a small number of studies were conducted specifically in EDs.

Chen et al. (8) conducted the first large-scale evaluation of Large Language Models (LLMs) for predicting a 30-day ED revisit using 138,010 visits from Stanford. They compared direct LLM predictions and embedding-based approaches against a traditional LightGBM model. While the generation did improve the LLMs performance, none of the models outperformed LightGBM, despite having access to both structured and unstructured clinical data. In particular, step by step reasoning had a reduction in performance in Claude 3.7, with analysis showing bias

such as overemphasising complex medical histories and overlooking protective factors. The study was conducted at a single US academic centre, therefore, its findings may not generalise to settings such as Caribbean EDS. Overall, the results suggest that advanced LLM reasoning does not necessarily improve clinical prediction and using conventional models with structured data may remain more reliable, especially in resource-limited settings.

Yang et al. (4), Adam et al. (5), Hinson et al. (6) and Rosenbacke et al. (7) all argue that how a model is built, what data it learns from and how it represents its outputs are as important as predictive accuracy. Boulitsakis Logothetis et al. (9) directly addresses this issue by demonstrating that interpretable models can achieve competitive accuracy without losing transparency. The study used electronic patient records from ED entry to compare Logistic Regression, Gradient Boosted Decision Trees and Support Vector Machines (SVM), a model that classifies data by drawing the widest possible boundary between categories. All three models performed well in predicting mortality in the hospital and clinical care admission, outperforming black-box alternatives. The study's primary limitation was its focus on a single NHS trust in the UK, limiting the applicability to clinical settings with different patient populations, practices or data systems.

While ML models can outperform traditional triage scales under controlled conditions, their reliability outside of training settings remains unverified Rockensschaub et al. (3), the data from which they learn contains biases that are unnoticed to the accuracy of metrics alone Yang et al. (4) and intentioned outputs can distort clinical judgement depending on how they are framed Adam et al. (5), Rosenbacke et al. (7). Critically, none of the studies reviewed were conducted in a Caribbean clinical setting or used data that reflected the patient population, resource limitations and documentation practices. This review identifies two gaps that the proposed pilot directly addresses: the lack of Caribbean validated AI triage and the need for output designs that maintain clinical practice in resource-limited settings.

Gap Analysis

Gap 1: Lack of Caribbean Validation

Porto. (1) reviewed 60 studies covering 57 ML algorithms and found that all validated models were developed using data from Asia, Europe or North America, with no study identifying or

validating an AI triage model on Caribbean patient data. This gap is clinically significant because Caribbean EDs serve distinct patient populations, frequently rely on paper-based systems and experience seasonal surges driven by visitor populations where conditions differ significantly from electronic settings in which existing models were trained. De Freitas et al. (10) qualitatively documented patient flow in a Caribbean ED and identified structural challenges including overcrowding, resource constraints and documentation practices that differ significantly from the electronic, high-resource settings in which existing ML models were developed. Validating an ML model using de-identified Mercer ED records would thus provide the first preliminary evidence of an AI-assisted triage performance in a Caribbean clinical setting.

Gap 2: Limited Evaluation of Output Design in Resource-Limited Settings

Model evaluation in the AI triage literature has primarily focused on predictive accuracy with little attention paid to how the outputs are designed for clinical use. Adam et al. (5) showed that framing AI outputs as descriptive risk flags better preserves independent clinical judgement, whereas Rosenbacke et al. (7) discovered that poorly calibrated trust in AI can cause clinicians to override sound judgement even when models are incorrect. Despite these findings, few studies have looked into how triage models should be structured to support rather than replace nurse-led decision making in resource limited settings. This pilot addresses that gap by comparing AI-generated triage recommendations against nurse-assigned categories using historical Mercer ED data with outputs purposely framed as descriptive risk flags to protect clinical autonomy.

Current ED Workflow and Stakeholders

To ensure that any proposed AI-assisted triage system aligns with existing Caribbean clinical practice, the current patient flow at Mercer General hospital was mapped using observations from Caribbean ED department workflows described by De Freitas et al. (10).

Patients arriving at the ED are initially screened for immediate life-threatening conditions. Patients who are classified by using the Emergency Severity Index (ESI) as a Level 1, bypass registration and are transferred directly to the resuscitation area for immediate treatment. All remaining patients proceed through registration before undergoing any triage assessment. During

triage, nurses collect patient history and records their vital signs. This includes, SpO2, blood pressure, temperature, heart rate and pain score. Following assessment, an ESI level (1-5) is assigned and patients are streamed according to acuity. High acuity patients are directed to the critical care area, while lower acuity patients are directed to fast-track or waiting areas.

Following streaming, ED clinicians perform a full assessment and request diagnostic investigations such as laboratory tests and imaging studies where required. Once investigations are completed, a disposition decision is made. Patients may be discharged, admitted to the ED ward or referred to an inpatient specialist team for review before transfer or discharge.

Figure 1. illustrates the complete triage workflow, highlighting operational constraints and potential AI integration points throughout the patient's journey.

Mercer General ED - Triage Process Flowchart

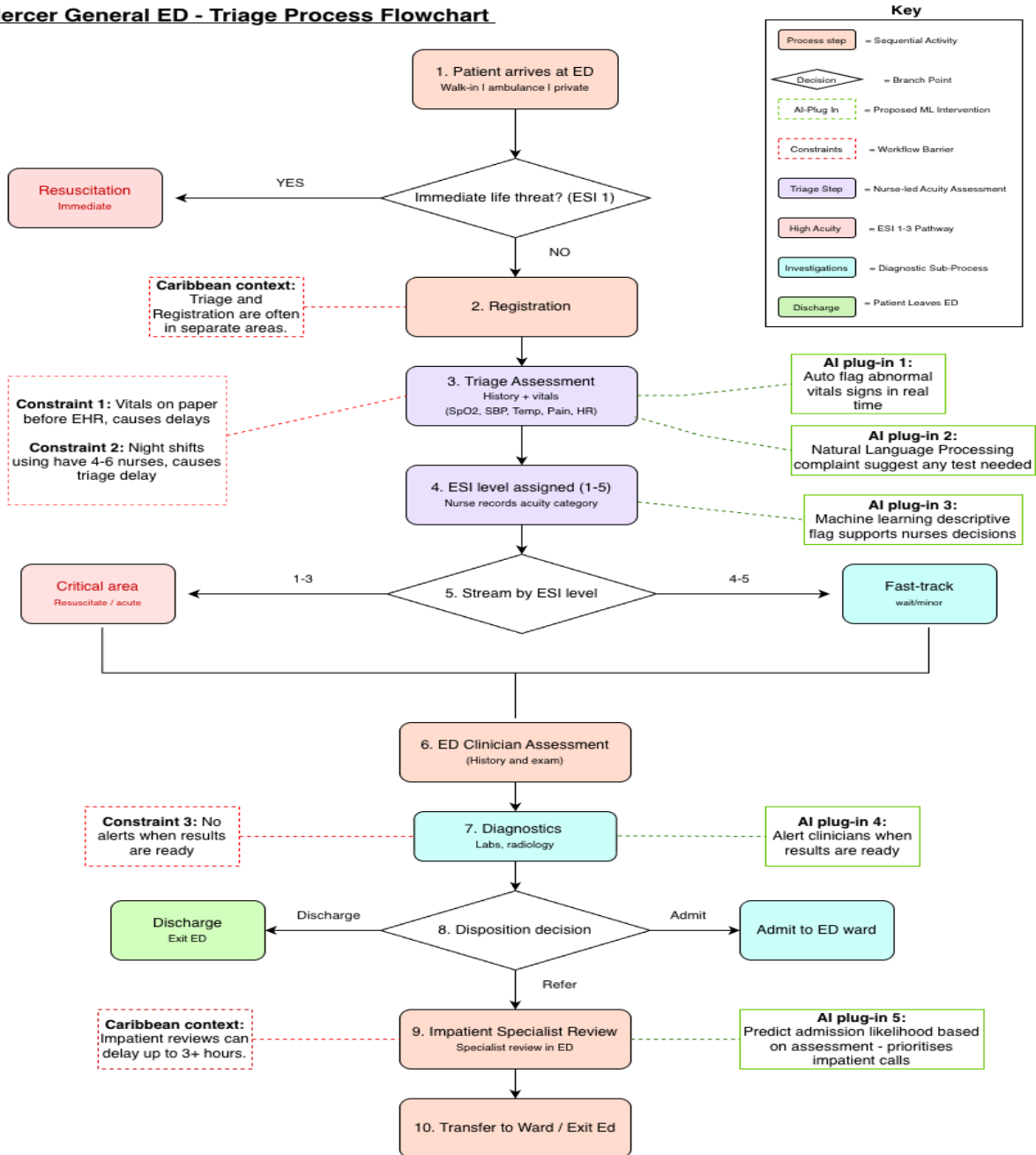


Figure 1. illustrates the complete ED triage workflow at Mercer General Hospital, from patient arrival and initial screening through triage, clinical assessment, diagnostics and final disposition. The diagram also highlights key operational constraints within the Caribbean context and identifies potential incorporation points for AI-assisted decision support across the patient pathway.

Key Stakeholders

Table 1. Stakeholder Analysis for the Proposed AI-Assisted Triage System

<i>Stakeholder</i>	<i>Role in Workflow</i>	<i>Interest in the Project</i>	<i>Key Benefits/Concerns</i>
<i>Patients</i>	Receive triage, assessment and treatment	Faster and more accurate care.	Reduced waiting times. Concerns about fairness and privacy
<i>Triage Nurses</i>	Assess patients, records vital signs and assigns ESI Levels	Reliable decision support without disturbing workflow.	Improved consistency. Concerns about added workload and loss of autonomy
<i>ED Clinicians</i>	Assess patients, order diagnostics and make disposition decisions	Better patient prioritisation and information access	Earlier identification of high risk patients Risk of overreliance on AI
<i>Inpatient Specialist</i>	Review patients requiring specialist care	Improved coordination and patient flow	Earlier identification of likely admissions. Concerns about inaccurate predictions
<i>Hospital Administration</i>	Oversees ED operations and resource allocation	Improved efficiency and service quality	Reduced overcrowding and better resource utilisation Concerns about implementation costs.
<i>IT Team</i>	Maintains HER and hospital information systems	Successful integration of AI tools	Improved digital capability Concerns about data quality, security and maintenance

Table 1. summarises the key stakeholders involved in the emergency department workflow and their interest in the proposed AI-assisted triage decision support system.

Workflow Constraints

Several operational constraints were identified within the current workflow and may influence the implementation and effectiveness of AI-assisted triage systems.

Constraint 1: Paper-Based Vital Sign Recording

Patient vitals are documented on paper before an Electronic Health Record (EHR) entry. This creates documentation delays and limits the effectiveness of AI models that rely on real-time clinical information.

Constraint 2: Nursing Shortages During Night Shifts

Staffing shortages frequently occur during night shifts, resulting in increased triage waiting times. De Freitas et al. (10) found about 4-6 nurses on shifts that required 14. Any AI system that requires additional manual data entry at the triage desk will be abandoned under those conditions.

Constraint 3: Lack of Diagnostic Results Notification

Diagnostic results are not automatically communicated to clinicians as identified by De Freitas et al. (10). Staff often need to manually retrieve reports, which contributes to delays in clinical decision making and patient flow.

Proposed Solution

This project proposes the development and validation of an AI-assisted triage decision support tool using de-identified data from Mercer General Hospital. The model will be trained on triage variables that are collected within a 3-5 minute window, which includes vital signs, pain score, mode of arrival and patient complaints. Rather than targeting fixed performance thresholds at this stage, the pilot aims to establish preliminary performance benchmarks and documents the direction and magnitude of any reduction in triage level inconsistencies in comparison to nurse-assigned CTAS and MTS categories. Given the 12-week timeframe these outcomes should be understood as exploratory rather than definitive, providing a foundation for future prospective evaluations.

Triage inconsistencies will be measured by comparing the model's recommended triage category against the category assigned by the triage nurse for each case. This approach directly addresses the gap identified in the literature, where model evaluation has focused on predictive accuracy without examining output agreement with clinical practice, Porto. (1)

To address the risk of algorithmic bias identified by Adam et al. (5), model outputs will be presented as descriptive flags rather than recommendations. For example, rather than having the model output "Assign ESI Level 2," the model would display a flag such as "High indicators present: elevated heart rate, pain score and ambulance arrival." This framing will keep clinical judgement central and reduce the risk of bias. Model outputs will be structured to be interpretable outlined by Boulitsakis Logothetis et al. (9) and address Sister Patrice Alleyne, Mercer's Director of Nursing, whose observation that clinical adoption remains Mercer's primary barrier to new tools.

Over the course of the 12-week training period, historical data will be used to train, validate and test the model. Concrete outputs will include a trained ML model, a performance evaluation report and a documented GitHub repository. These findings will provide the first preliminary evidence of AI-assisted triage in a Caribbean clinical setting and help guide the design of any future large-scale implementation across Caribbean EDs.

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Abbreviations and Key Terms

Abbreviation / Term	Definition / Description
AUROC	Area Under the Receiver: It is a performance metric used in machine learning and statistics to evaluate how well a classification model can distinguish between different classes
LightGBM	Light Gradient Boosting Machine: A highly effective gradient boosting framework that is optimised for performance and lowering memory usage
LLM	Large Language Model: advanced AI systems trained on massive datasets to read, understand, and generate human-like text.
NLP	Natural Language Processing: a branch of Artificial Intelligence (AI) that teaches computers to read, understand, and generate human language
PRISMA	Preferred Reporting Items for Systematic Review and Meta-Analyses: is a guideline designed to improve the reporting of systematic reviews. PRISMA provides authors with guidance and examples of how to completely report why a systematic review was done, what methods were used, and what results were found
Sentence Embedding	Multilingual Sentence Embedding: are mathematical representations of whole sentences. They help computers analyse and compare meanings by turning text into numeric values called vectors.
SVM	Support Vector Machine: is a supervised machine learning algorithm used for classification and regression tasks.
XGBoost	Extreme Gradient Boosting: is a distributed, open source machine learning library

XAI

Explainable Artificial Intelligence: is a set of processes and methods that allows human users to comprehend and trust the results and output created by machine learning algorithms.

Supplementary Information

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